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4th Sem / Electriacal

Subject : Utilization of Electrical Energy

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Throwing power can be improved by______. (CO4)

a) Decreasing distance between anode and cathode

b) Increasing distance between anode and cathode

c) Increase

d) None of these

Q.2 Electrodes used to direct resistance heating are made of (CO2)

a) Mild steel b) Copper

c) Brass d) Graphite

Q.3 The process of electrolysis is governed by ________. (CO3)

a) Faraday's Laws b) Joule's Laws

c) Ohm's Law d) Kirchhoff's Law

Q.4 Which type of motor is commonly used for industrial drives? (CO4)

a) Synchronous motor b) Induction motor

c) Stepper motor d) DC motor

Q.5 Which of the following is used in floodighting? (CO1)

a) Sodium vapour lamp

b) LED lamp

c) Incandescent lamp

d) Halogen lamp

Q.6 Resistance welding is suitable for____. (CO3)

a) Repair b) Production

c) Both (a) & (b) d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define Illumination. (CO1)

Q.8 In DC series motor, field winding is connected in____ with armature. (CO5)

Q.9 _____ lamps are widely used for street lighting. (CO1)

Q.10 In resistance heating, the heating element is generally made of_______. (CO2)

Q.11 Expand CFL. (CO1)

Q.12 A process used to protect metals from corrosion by coating them with a layer of zinc is called_____. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 State advantages of Electric heating process. (CO2)

Q.14 Write difference between Belt Drive and Chain drive. (CO5)

Q.15 Explain laws of electrolysis. (CO3)

Q.16 Explain inverse square law. (CO1)

Q.17 Why DC series Motor is best suited for traction work? (CO4)

Q.18 Draw and explain Mercury Vapour Lamp. (CO1)

Q.19 What are the laws of illumination flux and luminous efficiency? (CO1)

Q.20 Write characteristics of various types of mechanical loads. (CO4)

Q.21 Explain the process of dielectric heating and its applications. (CO2)

Q.22 Compare electric welding with other methods of welding. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain method of power transfer by using belt drive and chain drive. (CO4)

Q.24 Discuss factors influencing the selection of electric motors for industrial drives. (CO3)

Q.25 Explain series parallel control methods used for starting and speed control of traction motor. (CO5)